

Supplementary Material

Supplementary tables and figures for “When Does Opposition-Based Learning Actually Help? ...”. Numbers cross-reference the main text.

Supplementary Tables

Supplementary Table S1. Parameter settings of the compared algorithms (population-based algorithms: population size = 24; all: NFE_{max} = 500; 20 independent runs for the classical, origin-shifted, CEC2017, ablation, and factorial analyses; 15 independent runs for the far and mixed controls).

Algorithm	Parameter	Value	Source
IFPOA-X	Bandit exploration constant c	1.4	bandit_c
	Bandit reward signal	normalized HV gain	bandit_reward = "hv"
	Lévy stability index β	1.5	levy_beta
	Initial step scale c_0 (bounds)	0.15 ([0.01, 0.5])	c0 (c_min, c_max)
	Step anneal exponent η	1.5	eta_anneal
	1/5-rule target acceptance ρ^* (adapt factors)	0.20 ($\times 1.15$ / $\times 0.85$)	acc_target (c_adapt_delta, c_adapt_gamma)
	OBL trigger frequency	every 3 evaluations	obl_frequency
	OBL elite pool size	top-3 (front-1, by crowding)	obl_top_k
	JADE p-best fraction	20%	jade_p
	JADE F (Cauchy), CR (Normal)	Cauchy(0.5, 0.1), $\mathcal{N}(0.5, 0.1)$	jade_fmean, jade_cmean
	JADE external archive size	128	jade_arch_max
	Pareto archive size	64	archive_max
	k -NN surrogate / ASHA pruning	disabled for this benchmark	use_knn_screen = False (see §3.4)
	Switch probability p	0.8	p_switch
FPA	Local search scale γ	0.1	gamma
	Global (Lévy) step scale	0.25	levy_gamma
PSO	Inertia weight w	0.7	w
	Cognitive / social coefficients c_1, c_2	1.5, 1.5	c1, c2
DE	Mutation factor F (wf)	0.1 (<i>mealpy default, not tuned</i>)	mealpy DE.OriginalDE
	Crossover rate CR (cr)	0.9 (<i>mealpy default, not tuned</i>)	idem
	Strategy	rand/1/bin (strategy = 0)	idem
L-SHADE	Initial μ_F, μ_{CR}	0.5, 0.5 (<i>mealpy default, not tuned</i>)	mealpy SHADE.L_SHADE
	Population reduction	linear (L-SHADE default schedule)	idem
GWO	—	parameter-free by design (a : linear 2 $\rightarrow$ 0)	mealpy GWO.OriginalGWO
WOA	—	parameter-free by design (a : linear 2 $\rightarrow$ 0; spiral $b = 1$)	mealpy WOA.OriginalWOA
TPE	$n_{\text{startup_trials}}$	10 (<i>optuna default, not tuned</i>)	optuna.samplers.TPESampler
	$n_{\text{ei_candidates}}$	24 (<i>optuna default, not tuned</i>)	idem
	Prior weight, multivariate	1.0, off (<i>optuna default</i>)	idem

Supplementary Table S2. Summary statistics of final fitness (mean, D=50, 500 NFE, 20 runs). Best in bold.

Func	IFPOA-X	FPA	PSO	DE	L-SHADE	GWO	WOA	TPE
F1	7.38e+00	1.08e+04	8.41e+03	5.37e+04	2.13e+04	5.49e+03	3.18e+02	6.23e+04
F2	4.61e-01	6.08e+01	5.77e+01	5.37e+05	1.04e+05	4.89e+01	3.64e+00	1.91e+10
F3	1.17e+04	3.92e+04	3.28e+04	1.01e+05	9.96e+04	4.92e+04	3.26e+05	1.24e+05
F4	8.40e+00	3.31e+01	3.19e+01	7.21e+01	6.81e+01	4.54e+01	8.90e+01	7.61e+01
F5	8.42e+01	4.75e+06	2.92e+06	1.22e+08	2.38e+07	2.71e+06	2.23e+06	1.58e+08

Func	IFPOA-X	FPA	PSO	DE	L-SHADE	GWO	WOA	TPE
F6	8.20e+00	1.07e+04	8.67e+03	5.35e+04	2.20e+04	5.54e+03	1.76e+02	6.20e+04
F7	9.00e-02	4.06e+00	2.63e+00	8.69e+01	2.11e+01	2.36e+00	1.41e+00	1.13e+02
F8	1.59e+04	1.69e+04	1.54e+04	1.41e+04	1.64e+04	1.70e+04	6.29e+03	1.43e+04
F9	4.47e+00	4.17e+02	3.80e+02	4.98e+02	4.95e+02	4.25e+02	2.48e+02	5.68e+02
F10	6.38e-01	1.36e+01	1.31e+01	1.91e+01	1.69e+01	1.15e+01	2.08e+00	1.96e+01
F11	9.22e-01	9.85e+01	7.67e+01	4.84e+02	1.93e+02	5.04e+01	3.86e+00	5.62e+02
F12	1.40e+00	3.49e+05	1.58e+05	1.90e+08	1.82e+07	3.99e+05	2.12e+07	2.30e+08
F13	6.30e+00	6.02e+06	3.09e+06	4.15e+08	5.69e+07	1.55e+06	5.32e+06	5.52e+08
Mean rank	1.31	4.69	3.46	6.38	5.92	3.77	3.00	7.46

Supplementary Table S3. IFPOA-X vs baseline — per-function win/loss/tie (Mann–Whitney U, $\alpha = 0.05$, D=30). Per-function outcomes, not a cross-function significance claim (see §5.1).

Baseline	Win	Loss	Tie
FPA	13	0	0
PSO	12	0	1
DE	12	1	0
L-SHADE	13	0	0
GWO	13	0	0
WOA	12	1	0
TPE	12	1	0

Supplementary Table S4. Native-optimum verification of the shift transform (representative functions; full list in `verify_shift.py`). Residual is $|g(o)|$ at the shifted optimum; F7 is stochastic (noise term), so its residual reflects the noise floor.

Function	Native optimum x_0	Shifted optimum o (per-dim range)	min f	Verified $ g(o) $
F1 Sphere	0	30–70% of $[-100, 100]$	0	0.0e+00
F5 Rosenbrock	1	30–70% of $[-30, 30]$	0	0.0e+00
F8 Schwefel 2.26	420.97 · 1	in-bounds target	0	8.1e-09
F10 Ackley	0	30–70% of $[-32, 32]$	0	4.4e-16
F12 Penalized 1	−1	30–70% of $[-50, 50]$	0	3.6e-32
F13 Penalized 2	1	30–70% of $[-50, 50]$	0	1.4e-32

Complete statistical results promised in the main manuscript

All entries below are regenerated from the released raw CSV files. External inter-algorithm effects use independent run samples; ranks use the per-function medians.

Supplementary Table S5. CEC2017 final error: median [Q1, Q3] over 20 independent runs (D=30, 500 NFE).

Function	IFPOA-X	FPA	PSO	DE	L-SHADE	GWO	WOA	TPE
F1	4.062e+10 [3.853e+10, 4.665e+10]	5.437e+10 [4.715e+10, 6.051e+10]	2.803e+10 [2.409e+10, 3.322e+10]	5.142e+10 [4.053e+10, 5.895e+10]	2.162e+10 [1.835e+10, 2.415e+10]	9.577e+09 [7.111e+09, 1.270e+10]	5.666e+10 [5.179e+10, 6.500e+10]	4.244e+10 [3.724e+10, 4.666e+10]

Function	IFPOA-X	FPA	PSO	DE	L-SHADE	GWO	WOA	TPE
F3	8.068e+03 [6.392e+03, 9.694e+03]	1.186e+04 [9.624e+03, 1.441e+04]	4.488e+03 [3.195e+03, 5.765e+03]	9.902e+03 [7.698e+03, 1.207e+04]	3.592e+03 [2.690e+03, 3.940e+03]	1.335e+03 [8.138e+02, 1.889e+03]	1.868e+04 [1.424e+04, 2.002e+04]	6.370e+03 [5.194e+03, 7.983e+03]
F4	5.659e+04 [4.588e+04, 6.191e+04]	7.496e+04 [6.484e+04, 8.533e+04]	3.154e+04 [2.675e+04, 4.431e+04]	5.408e+04 [4.829e+04, 7.417e+04]	2.569e+04 [2.203e+04, 3.000e+04]	1.277e+04 [1.018e+04, 1.786e+04]	8.117e+04 [7.373e+04, 8.859e+04]	5.295e+04 [3.955e+04, 5.703e+04]
F5	5.314e-02 [4.414e-02, 6.599e-02]	4.316e-02 [3.479e-02, 5.114e-02]	2.436e-02 [2.109e-02, 3.373e-02]	3.259e-02 [2.479e-02, 4.367e-02]	5.752e-02 [4.703e-02, 6.310e-02]	2.363e-02 [1.474e-02, 4.678e-02]	5.170e-02 [4.159e-02, 6.438e-02]	3.111e-02 [2.078e-02, 3.485e-02]
F7	7.475e+02 [6.891e+02, 7.760e+02]	6.347e+02 [5.837e+02, 7.425e+02]	6.458e+02 [5.893e+02, 7.019e+02]	6.812e+02 [5.659e+02, 7.326e+02]	6.457e+02 [6.119e+02, 6.925e+02]	5.875e+02 [5.170e+02, 6.326e+02]	8.318e+02 [7.698e+02, 8.739e+02]	6.335e+02 [6.002e+02, 6.924e+02]
F9	8.305e+03 [7.893e+03, 8.598e+03]	7.951e+03 [7.681e+03, 8.383e+03]	7.574e+03 [7.250e+03, 7.875e+03]	8.371e+03 [7.992e+03, 8.660e+03]	8.626e+03 [8.296e+03, 8.901e+03]	8.283e+03 [8.161e+03, 8.731e+03]	7.839e+03 [7.410e+03, 8.063e+03]	8.440e+03 [8.025e+03, 8.642e+03]
F10	7.471e+08 [4.922e+08, 1.154e+09]	2.809e+09 [1.646e+09, 3.837e+09]	8.307e+07 [3.932e+07, 1.245e+08]	1.929e+09 [5.577e+08, 3.171e+09]	1.161e+08 [6.444e+07, 1.904e+08]	2.237e+06 [8.857e+05, 1.619e+07]	3.050e+09 [1.296e+09, 3.736e+09]	2.015e+08 [8.033e+07, 3.210e+08]
F11	5.014e+09 [4.562e+09, 5.695e+09]	1.154e+10 [8.322e+09, 1.486e+10]	1.786e+09 [1.291e+09, 2.212e+09]	5.714e+09 [3.863e+09, 7.725e+09]	1.738e+09 [1.504e+09, 2.221e+09]	3.251e+08 [2.390e+08, 5.845e+08]	9.944e+09 [8.075e+09, 1.307e+10]	2.756e+09 [1.865e+09, 3.504e+09]
F14	2.300e+09 [1.789e+09, 2.844e+09]	6.213e+09 [5.131e+09, 6.990e+09]	3.805e+08 [1.861e+08, 8.709e+08]	2.332e+09 [1.370e+09, 4.706e+09]	6.469e+08 [5.068e+08, 7.526e+08]	3.147e+07 [1.666e+07, 8.700e+07]	4.554e+09 [3.116e+09, 6.847e+09]	7.398e+08 [5.495e+08, 9.908e+08]
F21	9.015e+02 [7.276e+02, 1.058e+03]	2.542e+03 [2.301e+03, 2.985e+03]	6.869e+02 [6.037e+02, 8.620e+02]	1.313e+03 [9.916e+02, 2.136e+03]	4.705e+02 [4.387e+02, 5.432e+02]	3.407e+02 [3.065e+02, 3.978e+02]	3.092e+03 [2.200e+03, 4.215e+03]	9.018e+02 [8.134e+02, 9.890e+02]

Supplementary Table S6. CEC2017 mean Friedman ranks and 95% bootstrap confidence intervals (5000 resamples over functions).

Algorithm	Mean rank	95% CI
GWO	1.300	[1.00, 1.90]
PSO	2.700	[2.10, 3.30]
L-SHADE	3.600	[2.40, 5.20]
TPE	4.200	[3.40, 5.00]
IFPOA-X	5.300	[4.70, 5.90]
DE	5.700	[5.30, 6.00]
FPA	6.200	[5.00, 7.20]
WOA	7.000	[5.60, 7.80]

Supplementary Table S7. CEC2017 post-hoc comparisons against IFPOA-X (two-sided normal approximation to mean-rank differences).

Comparison	Delta rank	z	p raw	p Holm	p Finner
IFPOA-X vs GWO	+4.00	-3.65	2.607e-04	1.825e-03	1.824e-03
IFPOA-X vs PSO	+2.60	-2.37	1.762e-02	1.057e-01	6.033e-02
IFPOA-X vs L-SHADE	+1.70	-1.55	1.207e-01	6.035e-01	2.593e-01

Comparison	Delta rank	z	p raw	p Holm	p Finner
IFPOA-X vs TPE	+1.10	-1.00	3.153e-01	9.459e-01	4.116e-01
IFPOA-X vs DE	-0.40	+0.37	7.150e-01	9.459e-01	7.150e-01
IFPOA-X vs FPA	-0.90	+0.82	4.113e-01	9.459e-01	4.611e-01
IFPOA-X vs WOA	-1.70	+1.55	1.207e-01	6.035e-01	2.593e-01

Supplementary Table S8. CEC2017 independent-sample effect sizes (mean over functions).

Baseline	Vargha-Delaney A12	Mean Cohen's d
FPA	0.724	-1.05
PSO	0.132	+1.56
DE	0.556	-0.27
L-SHADE	0.158	+1.87
GWO	0.083	+2.89
WOA	0.799	-1.38
TPE	0.276	+0.94

Supplementary Table S9. Classical-suite mean Friedman ranks and 95% bootstrap confidence intervals (5000 resamples over functions).

Dimension	Algorithm	Mean rank	95% CI
D=30	IFPOA-X	1.308	[1.00, 1.92]
D=30	GWO	3.231	[2.62, 4.15]
D=30	WOA	3.385	[2.23, 4.69]
D=30	PSO	3.615	[3.15, 4.08]
D=30	FPA	4.692	[4.23, 5.23]
D=30	L-SHADE	6.000	[5.77, 6.23]
D=30	DE	6.385	[5.69, 6.92]
D=30	TPE	7.385	[6.38, 8.00]
D=50	IFPOA-X	1.308	[1.00, 1.92]
D=50	WOA	3.000	[1.92, 4.31]
D=50	PSO	3.462	[3.00, 3.85]
D=50	GWO	3.769	[3.15, 4.62]
D=50	FPA	4.692	[4.08, 5.31]
D=50	L-SHADE	5.923	[5.77, 6.00]
D=50	DE	6.385	[5.46, 7.00]
D=50	TPE	7.462	[6.62, 8.00]

Supplementary Table S10. Classical-suite post-hoc comparisons against IFPOA-X.

Dimension	Comparison	Delta rank	z	p raw	p Holm	p Finner
D=30	IFPOA-X vs GWO	-1.92	+2.00	4.533e-02	6.128e-02	4.533e-02
D=30	IFPOA-X vs WOA	-2.08	+2.16	3.064e-02	6.128e-02	3.565e-02
D=30	IFPOA-X vs PSO	-2.31	+2.40	1.631e-02	4.893e-02	2.276e-02
D=30	IFPOA-X vs FPA	-3.38	+3.52	4.270e-04	1.708e-03	7.471e-04
D=30	IFPOA-X vs L-SHADE	-4.69	+4.88	1.040e-06	5.200e-06	2.427e-06

Dimension	Comparison	Delta rank	z	p raw	p Holm	p Finner
D=30	IFPOA-X vs DE	-5.08	+5.28	1.262e-07	7.574e-07	4.418e-07
D=30	IFPOA-X vs TPE	-6.08	+6.33	2.531e-10	1.772e-09	1.772e-09
D=50	IFPOA-X vs WOA	-1.69	+1.76	7.817e-02	7.817e-02	7.817e-02
D=50	IFPOA-X vs PSO	-2.15	+2.24	2.497e-02	4.995e-02	2.908e-02
D=50	IFPOA-X vs GWO	-2.46	+2.56	1.041e-02	3.122e-02	1.454e-02
D=50	IFPOA-X vs FPA	-3.38	+3.52	4.270e-04	1.708e-03	7.471e-04
D=50	IFPOA-X vs L-SHADE	-4.62	+4.80	1.556e-06	7.782e-06	3.632e-06
D=50	IFPOA-X vs DE	-5.08	+5.28	1.262e-07	7.574e-07	4.418e-07
D=50	IFPOA-X vs TPE	-6.15	+6.41	1.502e-10	1.052e-09	1.052e-09

Supplementary Table S11. Classical-suite independent-sample effect sizes (mean over functions).

Dimension	Baseline	Vargha-Delaney A12	Mean Cohen's d
D=30	FPA	0.980	-4.90
D=30	PSO	0.946	-5.12
D=30	DE	0.925	-7.76
D=30	L-SHADE	0.979	-8.27
D=30	GWO	0.993	-3.88
D=30	WOA	0.877	-2.10
D=30	TPE	0.924	-10.33
D=50	FPA	0.983	-6.73
D=50	PSO	0.936	-5.42
D=50	DE	0.924	-8.16
D=50	L-SHADE	0.988	-9.12
D=50	GWO	0.983	-4.46
D=50	WOA	0.848	-1.83
D=50	TPE	0.924	-11.66

Supplementary Figures

Supplementary Figure S1. Rank distribution per (function, run) for each algorithm (D=30).

Supplementary Figure S2. Mean rank (Friedman) per algorithm, D=30 vs D=50.

Implementation Details

Computational complexity

Per iteration, IFPOA-X performs: one incumbent (re-)evaluation subject to $O(1)$ amortized cost via memoized caching on (θ, rung) ; a JADE mutation in $O(D)$; an OBL trigger every 3 iterations costing $O(D)$ for the opposite computation plus $O(|\mathcal{F}_1| \log |\mathcal{F}_1|)$ for non-dominated sorting and crowding-distance ranking of the elite pool, where $|\mathcal{F}_1|$ is the size of the current Pareto front; a Pareto-archive update in $O(|\mathcal{A}|)$ (or $O(|\mathcal{A}| \log |\mathcal{A}|)$) when trimming by crowding distance once the archive exceeds `archive_max` = 64; and an $O(1)$ bandit index update. When the optional k -NN screen is enabled (disabled in this paper's experiments, §3.4), each screened candidate adds $O(N \cdot D)$ for a brute-force nearest-neighbour query over N stored surrogate points. Overall, the per-iteration overhead beyond the objective evaluation itself is dominated by $O(D + |\mathcal{A}| \log |\mathcal{A}|)$, which is negligible relative to the cost of a single real evaluation in the expensive-optimization regime this paper targets — precisely the assumption that justifies trading a modest bookkeeping overhead for improved sample efficiency.

Full algorithm (pseudocode)

Algorithm S1. IFPOA-X (single-objective instantiation used in this paper)

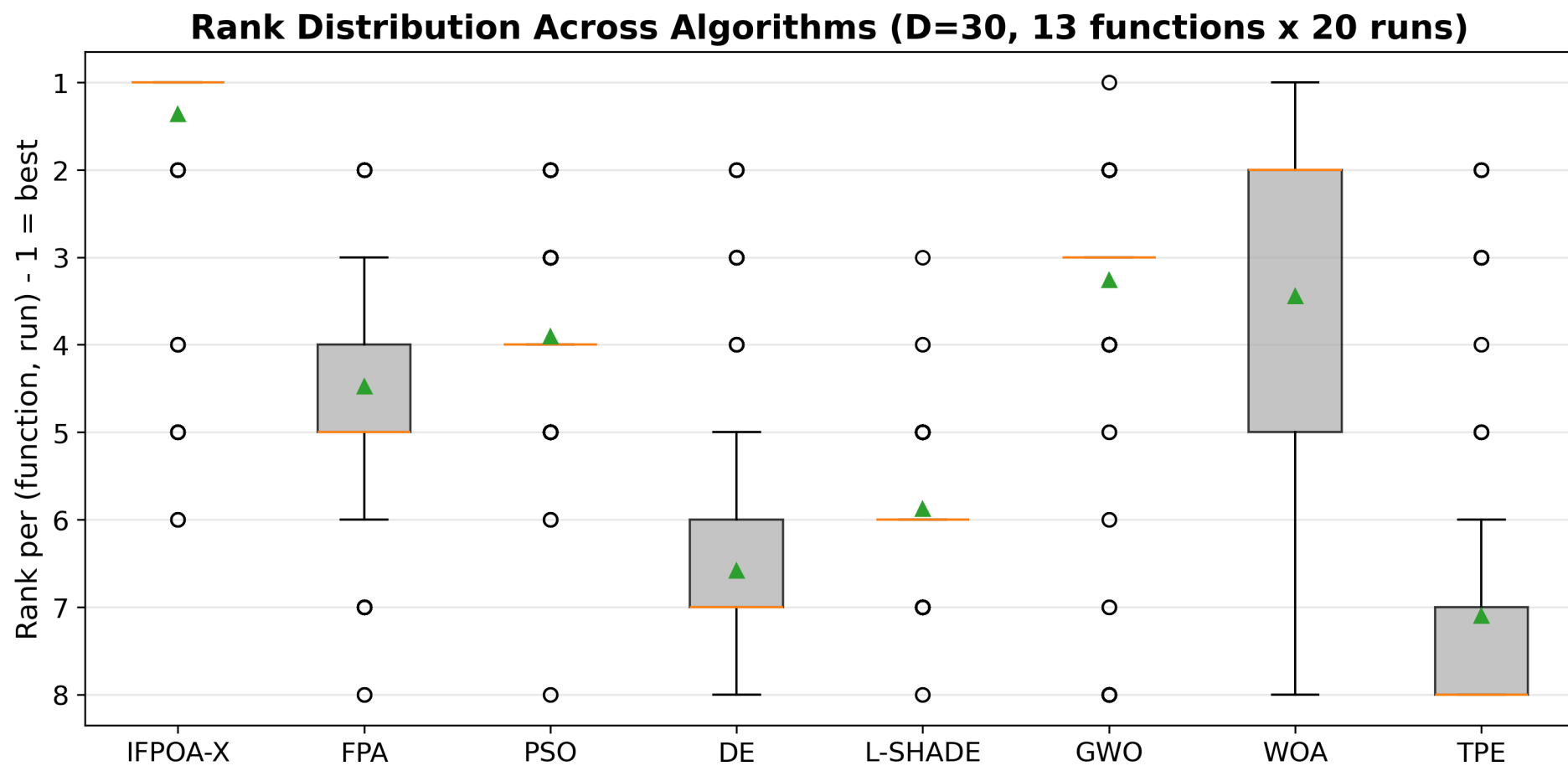


Figure 1: Rank boxplot

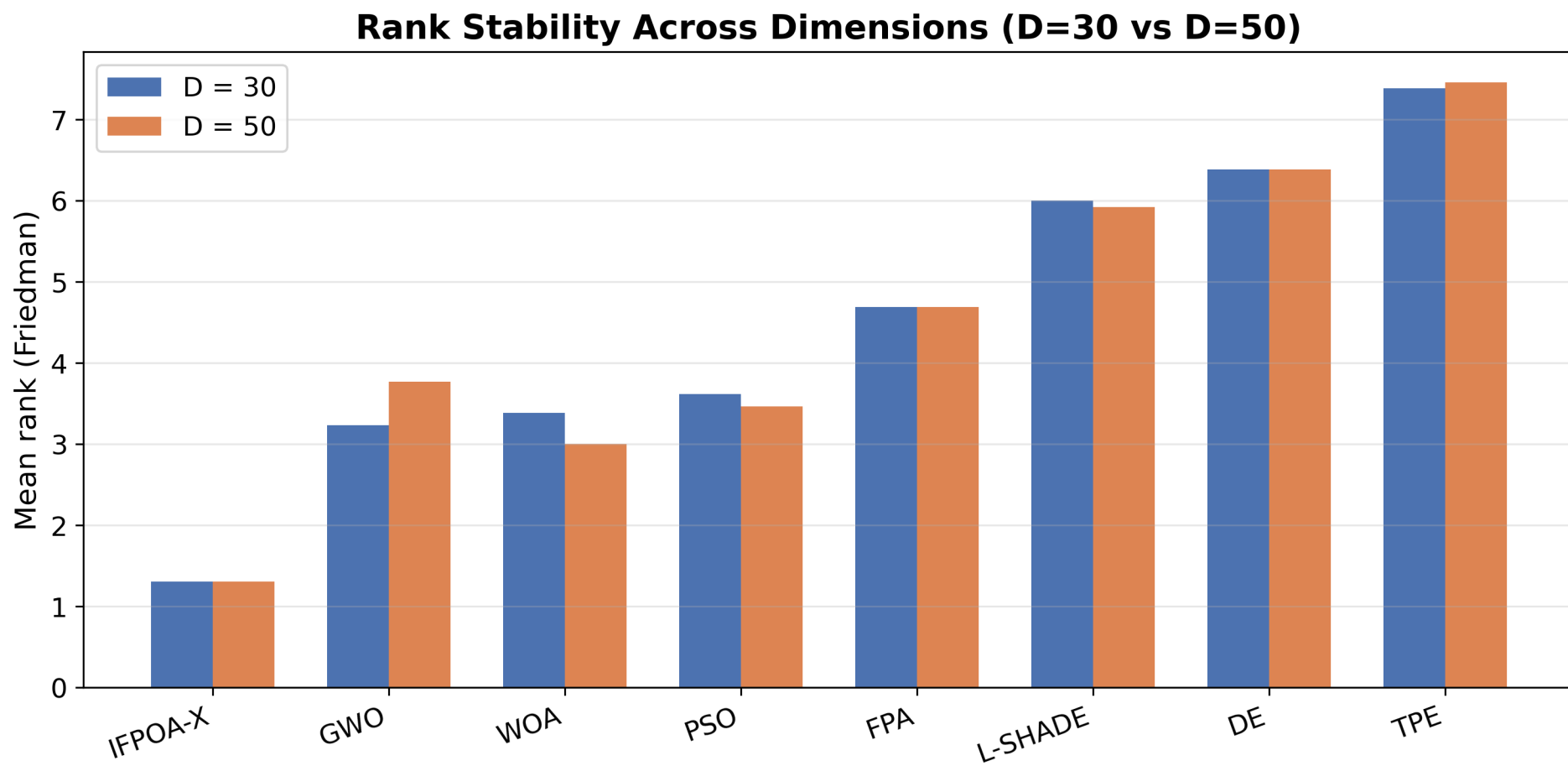


Figure 2: Scalability

Input: budget NFE_max, population size N, dimension D

Output: best solution found

```
1: Initialize population {u_1, ..., u_N} via Latin Hypercube Sampling in [0,1]^D
2: t <- 0; archive <- empty; bandit stats <- {N_global=0, N_local=0, R_global=0, R_local=0}
3: while t < NFE_max do
4:   idx <- select an unevaluated (or stalest-evaluated) individual
5:   fit <- Evaluate(u_idx) // real evaluation (or cache hit)
6:   t <- t + 1
7:   Update Pareto archive and best-so-far with (u_idx, fit)
8:   if t mod obl_frequency == 0 then
9:     u_elite <- random draw from top-k elite pool (archive front-1, by crowding)
10:    u_opp <- 1 - u_elite // OBL opposite
11:    fit_opp <- Evaluate(u_opp); t <- t + 1 // consumes 1 NFE
12:    if u_opp not dominated by cohort-worst then replace cohort-worst with u_opp
13:  end if
14:  a <- SelectArm_UCB1(bandit stats, c=1.4) // "global" or "local"
15:  c_t <- CalibrateStepSize(t, accept_history) // anneal + 1/5-rule
16:  if a == "global" then
17:    u_child <- clip(u_idx + c_t * LevyStep(beta=1.5), 0, 1)
18:  else
19:    F ~ Cauchy(0.5, 0.1); CR ~ Normal(0.5, 0.1)
20:    u_pbest <- sample from top-20% of archive front-1
21:    u_r1, u_r2 <- sample distinct from population union external archive
22:    u_child <- CurrentToPBest1(u_idx, u_pbest, u_r1, u_r2, F, CR) // JADE
23:  end if
24:  [optional: if surrogate screen enabled, discard u_child if predicted-unpromising]
25:  fit_child <- Evaluate(u_child); t <- t + 1
26:  Update Pareto archive and best-so-far with (u_child, fit_child)
27:  reward <- clip((HV_after - HV_before) / A_ref, 0, 1)
28:  UpdateBandit(a, reward)
29: end while
30: return best-so-far
```